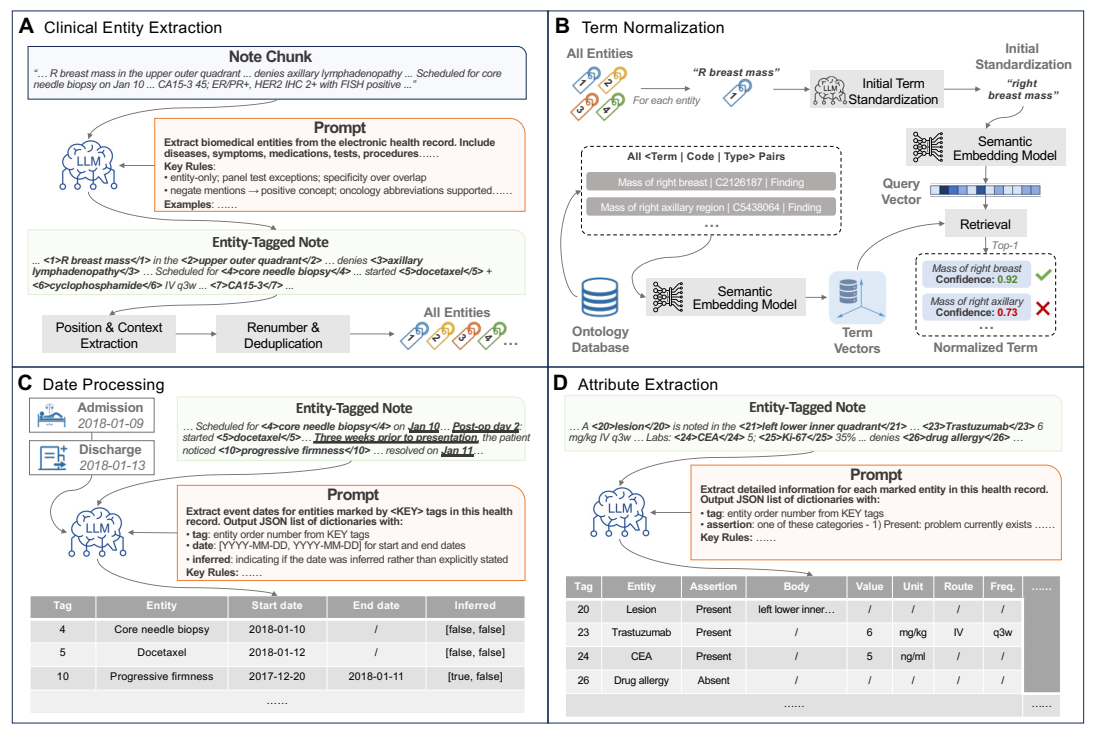

A Clinical Entity Extraction
Note Chunk
"… R breast mass in the upper outer quadrant … denies axillary lymphadenopathy … Scheduled for core needle biopsy on Jan 10 … CA15-3 45; ER/PR+, HER2 IHC 2+ with FISH positive …"
LLM
Prompt
Extract biomedical entities from the electronic health record. Include diseases, symptoms, medications, tests, procedures……
Key Rules:
• entity-only; panel test exceptions; specificity over overlap
• negate mentions → positive concept; oncology abbreviations supported……
Examples: ……
Entity-Tagged Note
… <1>R breast mass</1> in the <2>upper outer quadrant</2> … denies <3>axillary lymphadenopathy</3> … Scheduled for <4>core needle biopsy</4> … started <5>docetaxel</5> + <6>cyclophosphamide</6> IV q3w … <7>CA15-3</7> …
Position & Context Extraction
Renumber & Deduplication
All Entities
B Term Normalization
All Entities
For each entity
"R breast mass"
LLM
Initial Term Standardization
Initial Standardization
"right breast mass"
Semantic Embedding Model
All <Term | Code | Type> Pairs
Mass of right breast | C2126187 | Finding
Mass of right axillary region | C5438064 | Finding
Query Vector
Retrieval
Top-1
Ontology Database
Semantic Embedding Model
Term Vectors
Mass of right breast
Confidence: 0.92
Mass of right axillary
Confidence: 0.73
Normalized Term
C Date Processing
Admission
2018-01-09
Discharge
2018-01-13
Entity-Tagged Note
… Scheduled for <4>core needle biopsy</4> on Jan 10 … Post-op day 2: started <5>docetaxel</5> … Three weeks prior to presentation, the patient noticed <10>progressive firmness</10> … resolved on Jan 11 …
LLM
Prompt
Extract event dates for entities marked by <KEY> tags in this health record. Output JSON list of dictionaries with:
• tag: entity order number from KEY tags
• date: [YYYY-MM-DD, YYYY-MM-DD] for start and end dates
• inferred: indicating if the date was inferred rather than explicitly stated
Key Rules: ……
Tag | Entity | Start date | End date | Inferred
4 | Core needle biopsy | 2018-01-10 | / | [false, false]
5 | Docetaxel | 2018-01-12 | / | [false, false]
10 | Progressive firmness | 2017-12-20 | 2018-01-11 | [true, false]
……
D Attribute Extraction
Entity-Tagged Note
… A <20>lesion</20> is noted in the <21>left lower inner quadrant</21> … <23>Trastuzumab</23> 6 mg/kg IV q3w … Labs: <24>CEA</24> 5; <25>Ki-67</25> 35% … denies <26>drug allergy</26> …
LLM
Prompt
Extract detailed information for each marked entity in this health record. Output JSON list of dictionaries with:
• tag: entity order number from KEY tags
• assertion: one of these categories - 1) Present: problem currently exists ……
Key Rules: ……
Tag | Entity | Assertion | Body | Value | Unit | Route | Freq. | ……
20 | Lesion | Present | left lower inner… | / | / | / | / |
23 | Trastuzumab | Present | / | 6 | mg/kg | IV | q3w |
24 | CEA | Present | / | 5 | ng/ml | / | / |
26 | Drug allergy | Absent | / | / | / | / | / |
……